

ARTIFICIAL INTELLIGENCE LAB

BCSE AIML 2024 SECTION C

PROCESS OF THE LAB

Open Google Colab



Write the AI program



Run the code



Save the notebook (.ipynb)



Save/Upload it to GitHub

GITHUB

1. In Colab, click File → Download → Download .ipynb.
2. Open the GitHub repository in a browser.
3. Click Add file → Upload files.
4. Drag and drop the downloaded notebook.
5. Add a commit message.
6. Click Commit changes.

Experiment 1: Introduction to **Python Libraries for Artificial** **Intelligence**

Library	Demonstration
• NumPy	Create arrays and perform arithmetic
• Pandas	Create a DataFrame
• Matplotlib	Plot a graph
• Scikit-learn	Train a simple Linear Regression model
• TensorFlow	Create a tensor and perform addition
• OpenCV	Draw a circle or rectangle on an image
• NLTK	Tokenize a sentence
• Seaborn (optional)	Display a heatmap
• PIL	Open and display an image



AI Libraries Demonstration



NumPy



Pandas



Matplotlib



Scikit-learn



TensorFlow



OpenCV



NLTK



Conclusion

DEMONSTRATION

```
=====
# AI LAB EXPERIMENT 1
```

```
# Introduction to Python Libraries for Artificial Intelligence
=====
```

```
# Import Libraries
```

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import tensorflow as tf
```

```
import cv2
```

```
from sklearn.preprocessing import LabelEncoder
```

```
print("=" * 60)
```

```
print("INTRODUCTION TO PYTHON LIBRARIES FOR AI")
```

```
print("=" * 60)
```

```
# =====
```

```
# NUMPY
```

```
# =====
```

```
print("\n----- NumPy -----")
```

```
numbers = np.array([10, 20, 30, 40, 50])
```

```
print("Array :", numbers)
```

```
print("Sum :", np.sum(numbers))
```

```
print("Mean :", np.mean(numbers))
```

```
print("Maximum :", np.max(numbers))
```

```
# =====  
# PANDAS  
# =====  
  
print("\n----- Pandas -----")  
  
student = {  
    "Name": ["Alice", "Bob", "Charlie"],  
    "Marks": [85, 90, 78]  
}  
  
df = pd.DataFrame(student)  
  
print("\nStudent Data")  
print(df)
```

#

=====

MATPLOTLIB

#

=====

print("\n----- Matplotlib -----")

print("Displaying Bar Chart...")

plt.figure(figsize=(6,4))

plt.bar(df["Name"], df["Marks"])

plt.title("Student Marks")

plt.xlabel("Student Name")

plt.ylabel("Marks")

plt.show()

```
# =====  
# SCIKIT-LEARN  
# =====  
  
print("\n----- Scikit-learn -----")  
  
encoder = LabelEncoder()  
  
df["Student ID"] = encoder.fit_transform(df["Name"])  
  
print("After Encoding Names into Numbers")  
print(df)
```

```
# =====
```

```
# TENSORFLOW
```

```
# =====
```

```
print("\n----- TensorFlow -----")
```

```
tensor1 = tf.constant([1, 2, 3])
```

```
tensor2 = tf.constant([4, 5, 6])
```

```
print("Tensor 1 :", tensor1.numpy())
```

```
print("Tensor 2 :", tensor2.numpy())
```

```
result = tf.add(tensor1, tensor2)
```

```
print("Tensor Addition :", result.numpy())
```

```
# =====  
# OpenCV  
# =====  
  
print("\n----- OpenCV -----")  
# Create a blank black image  
image = np.zeros((250,250,3), dtype=np.uint8)  
# Draw a blue rectangle  
cv2.rectangle(image, (50,50), (200,200), (255,0,0), 3)  
# Draw a green circle  
cv2.circle(image, (125,125), 40, (0,255,0), 3)  
# Display image  
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))  
plt.title("OpenCV Shapes")  
plt.axis("off")  
plt.show()
```

=====

CONCLUSION

=====

print("\n" + "=" * 60)

print("Experiment Completed Successfully")

print("=" * 60)

print("\nLibraries Demonstrated:")

print("NumPy")

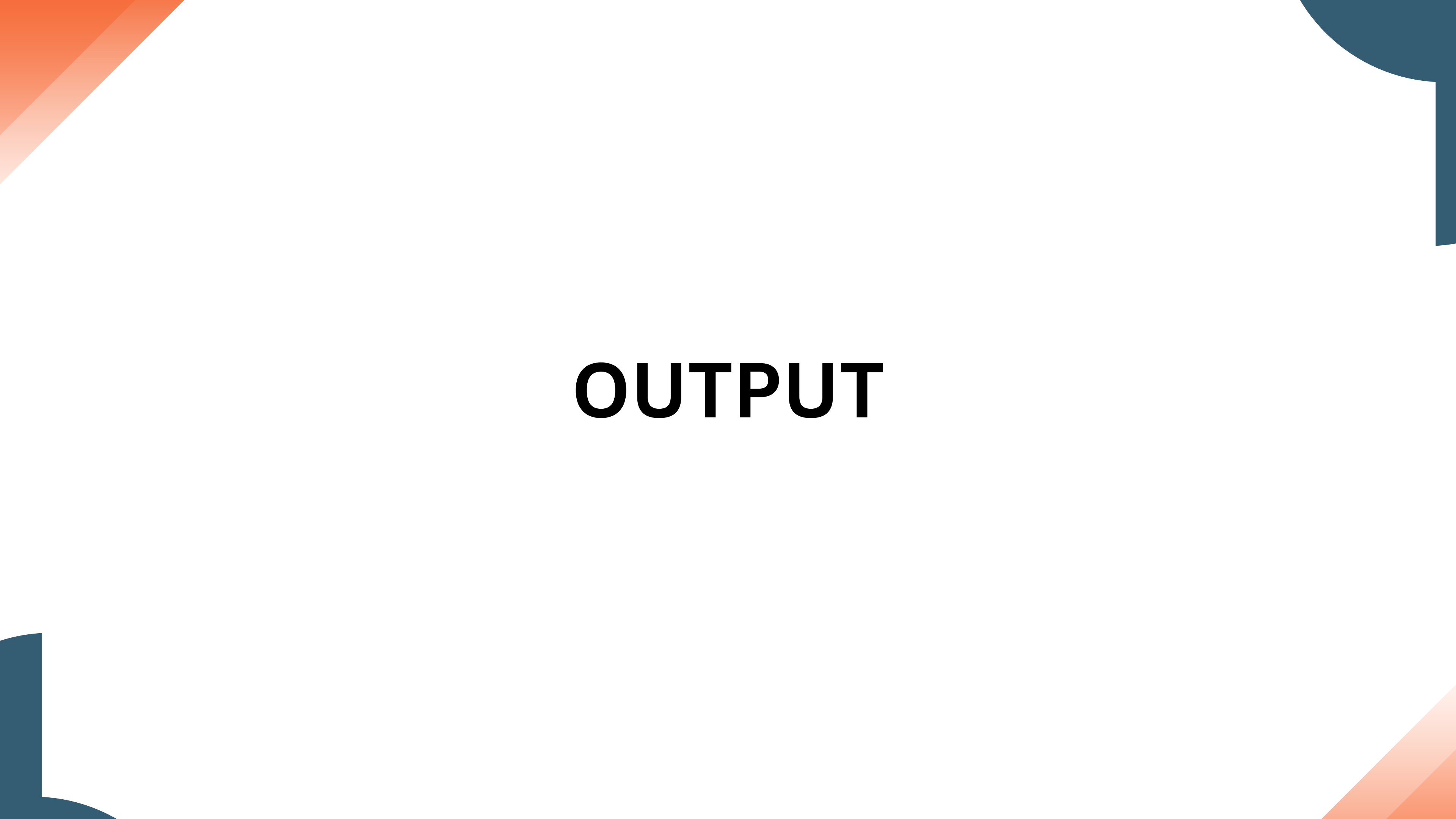
print("Pandas")

print("Matplotlib")

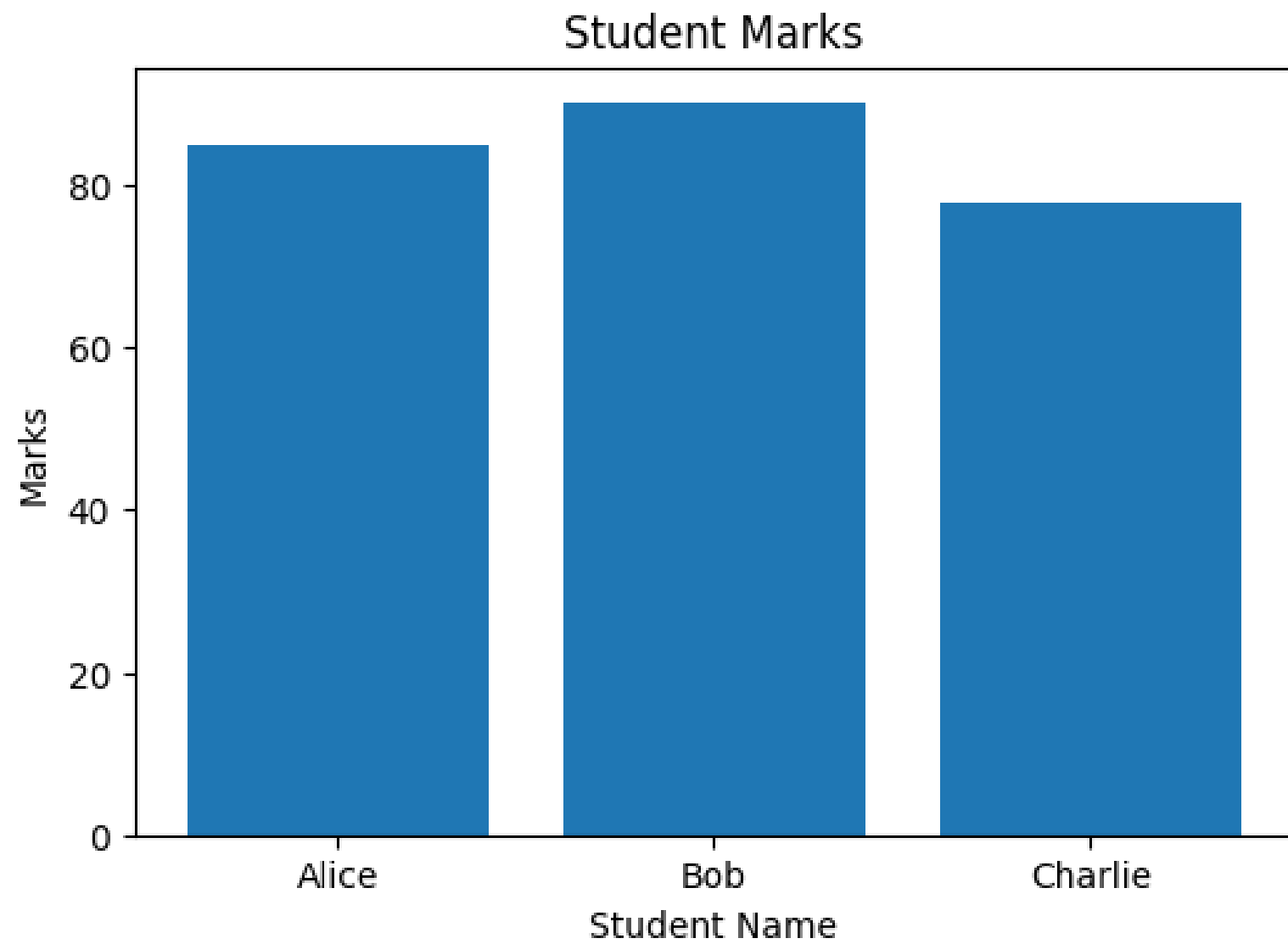
print("Scikit-learn")

print("TensorFlow")

print("OpenCV")



OUTPUT



----- Scikit-learn -----

After Encoding Names into Numbers

Name Marks Student ID

0 Alice 85 0

1 Bob 90 1

2 Charlie 78 2

----- TensorFlow -----

Tensor 1 : [1 2 3]

Tensor 2 : [4 5 6]

Tensor Addition : [5 7 9]

OpenCV Shapes

